DIG5127 Database and Web Application Development

Milestone 2 - ERM proposal

Author: Samikshya Dhamala

Student Id: 25123833

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Product Stock Management System

1. Introduction

Inventory management is an important aspect of businesses that involve the acquisition, storage, and sale of tangible products. This is because ineffective management of products in inventory can result in problems such as loss of stock, overstocking, expired products, and lost revenue. This project will involve the development of a "Product Stock Management System" in the business that operates as an intermediary by purchasing products from wholesalers to sell to clients.

Overall, this Product Stock Management System aims to improve operational efficiency, ensure accurate stock control, and reduce manual errors by providing a structured and reliable inventory management solution suitable for small to medium-scale businesses.

2. Application Overview

The system is aimed at handling products from buying the products to selling them. It keeps records of all products, suppliers, purchase orders, and even sale orders. To make it more realistic, it keeps track of items that come in batches, especially those that are consumable.

The system enables the business to generate purchase orders for supplies, store items received in batches, and update the level of available items automatically. In relation to sales, items are sold from the available batches, ensuring that stock is deducted correctly, as well as managing items that expire. There is also an indication of the current level of stock.

3. Functionality

User Management
- Users are internal staff who log in to the system

Supplier Management
- Add, edit, delete suppliers
- Map products to suppliers

Product Management
- Add, edit, delete products
- Monitor stock and low-stock alerts

Purchase Management
- Create purchase orders with items
- Record batches with expiry dates
- Update stock automatically

Sales Management
- Record sales and sale items
- Deduct sold quantity from stock and batches
- Record payments

Stock Management

- Track product quantity and batch availability
- Manage expiry and reorder levels

4. Entity Relationship Model (ERM)

1. user_owner

- Purpose: Stores system users (admins and normal users) who can log in.

Attribute	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each user
username	VARCHAR(255)	NOT NULL, UNIQUE	Name of the user
password	VARCHAR(255)	NOT NULL	Hashed password
email	VARCHAR(255)	NOT NULL, UNIQUE	Email address
role	ENUM('admin', 'user')	NOT NULL, DEFAULT 'user'	Role type of the user
status	ENUM('active', 'inactive')	NOT NULL, DEFAULT 'active'	Account status

2. user_otp

- Purpose: Temporary OTP storage for registration or password reset.

Attribute	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique OTP record ID
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES user(user_id)	Links to the user owning the OTP
otp	VARCHAR(6)	NOT NULL	One-time password
expires_at	DATETIME	NOT NULL	Expiry time of the OTP
is_used	TINYINT(1)	NOT NULL, DEFAULT 0	Flag indicating if OTP has been used (0 = no, 1 = yes)
no_of_attempts	INT	NOT NULL, DEFAULT 0	Number of attempts made for this OTP

purpose	ENUM('register', 'reset')	NOT NULL	Purpose of OTP (registration or password reset)

3. supplier

- Purpose: Stores supplier information for purchase orders.

Attribute	Data Type	Constraints	Description
supplier_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each supplier
supplier_name	VARCHAR(255)	NOT NULL	Name of the supplier
supplier_address	TEXT	NULL	Address of the supplier
supplier_phone	VARCHAR(20)	NULL	Contact number of the supplier
supplier_email	VARCHAR(255)	NULL, UNIQUE	Email address of the supplier
contact_person	VARCHAR(255)	NULL	Name of the contact person at supplier
status	ENUM('active', 'inactive')	NOT NULL, DEFAULT 'active'	Status of the supplier
created_at	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date the supplier was added

4. product

- Purpose: Stores all product details.

Attribute	Data Type	Constraints	Description
product_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each product
product_name	VARCHAR(255++)	NOT NULL	Name of the product

product_category	VARCHAR(255)	NULL	Category of the product
unit	VARCHAR(50)	NOT NULL	Unit type (kg, pcs, liter, etc.)
selling_price	DECIMAL(10,2)	NOT NULL	Selling price per unit
reorder_level	INT	NOT NULL, DEFAULT 0	Minimum stock level before reorder
status	ENUM('active','inactive')	NOT NULL, DEFAULT 'active'	Product status
quantity	INT	NOT NULL, DEFAULT 0	Total quantity available

5. purchase_order

- Purpose: Tracks orders placed by users to suppliers.

Attribute	Data Type	Constraints	Description
purchase_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique purchase order ID
supplier_id	INT	NOT NULL, FOREIGN KEY REFERENCES supplier(supplier_id)	Linked supplier for this order
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES user(user_id)	User who placed the order
order_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date the order was placed
expected_delivery	DATE	NULL	Expected delivery date

status	ENUM('pending','completed','cancelled')	NOT NULL, DEFAULT 'pending'	Status of the purchase order

6. purchased_item

- Purpose: Stores items included in each purchase order.

Attribute	Data Type	Constraints	Description
purchase_item_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each purchased item
purchase_id	INT	NOT NULL, FOREIGN KEY REFERENCES purchase(purchase_id)	Links item to the purchase order
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES product(product_id)	Product that was purchased
quantity_order	INT	NOT NULL	Quantity ordered for the product
unit_price	DECIMAL(10,2)	NOT NULL	Price per unit at time of purchase

7. batch

- Purpose: Tracks batch-level inventory of products.

Attribute	Data Type	Constraints	Description
batch_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each batch
batch_no	VARCHAR(100)	NOT NULL	Batch number assigned by supplier/manufacturer
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES product(product_id)	Product linked to this batch

purchase_id	INT	NOT NULL, FOREIGN KEY REFERENCES purchase(purchase_id)	Purchase order associated with this batch
manufacture_date	DATE	NULL	Manufacturing date of the product batch
expire_date	DATE	NULL	Expiry date of the product batch
quantity_received	INT	NOT NULL	Total quantity received in this batch
quantity_remaining	INT	NOT NULL	Quantity left in stock from this batch

8. sale

- Purpose: Stores sales transactions.

Attribute	Data Type	Constraints	Description
sale_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each sale
sale_date	DATE	NOT NULL, DEFAULT CURRENT_DATE	Date the sale was made
total_amount	DECIMAL(10,2)	NOT NULL	Total sale amount
payment	VARCHAR(50)	NOT NULL	Payment method (cash, card, online, etc.)
status	ENUM('completed','pending','cancelled')	NOT NULL, DEFAULT 'completed'	Status of the sale

9. sale_item

- Purpose: Stores items sold in a sale transaction.

Attribute	Data Type	Constraints	Description
sale_item_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique ID for each sold item
sale_id	INT	NOT NULL, FOREIGN KEY REFERENCES sale(sale_id)	Links item to the sale transaction
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES product(product_id)	Product that was sold
batch_id	INT	NOT NULL, FOREIGN KEY REFERENCES batch(batch_id)	Batch from which the product was sold
quantity_sold	INT	NOT NULL	Quantity sold from the batch
sale_price	DECIMAL(10,2)	NOT NULL	Price per unit at which the product was sold

5. Relationships

- user_owner and user_otp
  Type: One-to-Many (1:N)
  Details: One user can have multiple OTP entries for actions like registration or password reset.

- user_owner and purchase_order
  Type: One-to-Many (1:N)
  Details: One user (buyer) can place multiple purchase orders.

- supplier and purchase_order
  Type: One-to-Many (1:N)
  Details: One supplier can supply multiple purchase orders.

- purchase_order and purchased_item
  Type: One-to-Many (1:N)
  Details: One purchase order can include multiple products (items).

- product and purchased_item
  Type: One-to-Many (1:N)
  Details: One product can appear in multiple purchased items.

- purchase_order and batch
  Type: One-to-Many (1:N)

Details: One purchase order can generate multiple batches of products.

- product and batch
  Type: One-to-Many (1:N)
  Details: One product can have multiple batches.

- Sale and sale_item
  Type: One-to-Many (1:N)
  Details: One sale transaction can include multiple sale items.

- product and sale_item
  Type: One-to-Many (1:N)
  Details: One product can appear in multiple sale items.

- batch and sale_item
  Type: One-to-Many (1:N)
  Details: Each batch can be sold across multiple sale items (tracking batch-wise sales).

6. Entity Relationship Model (ERM) Diagram

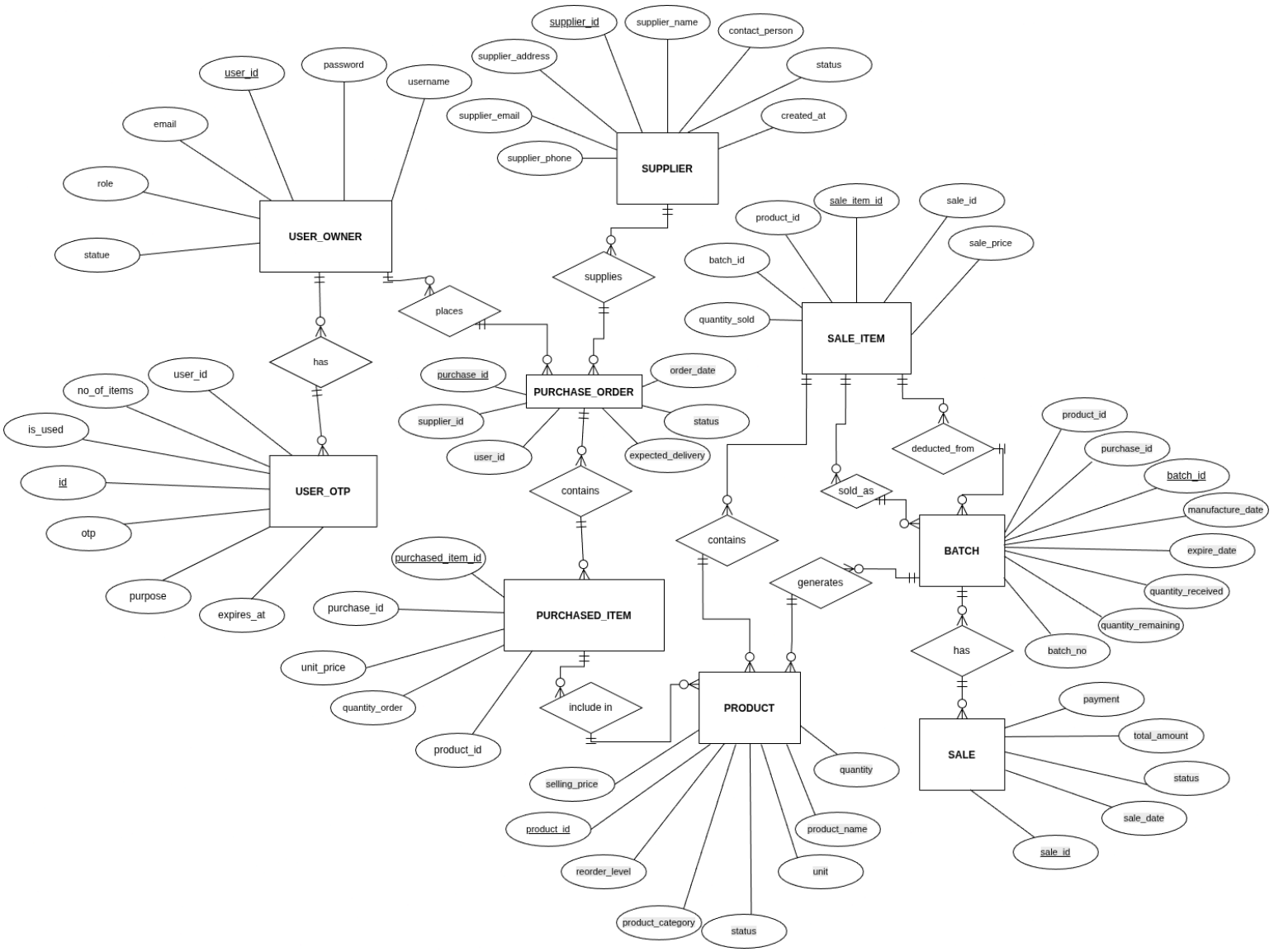

Fig I: ERM of my system

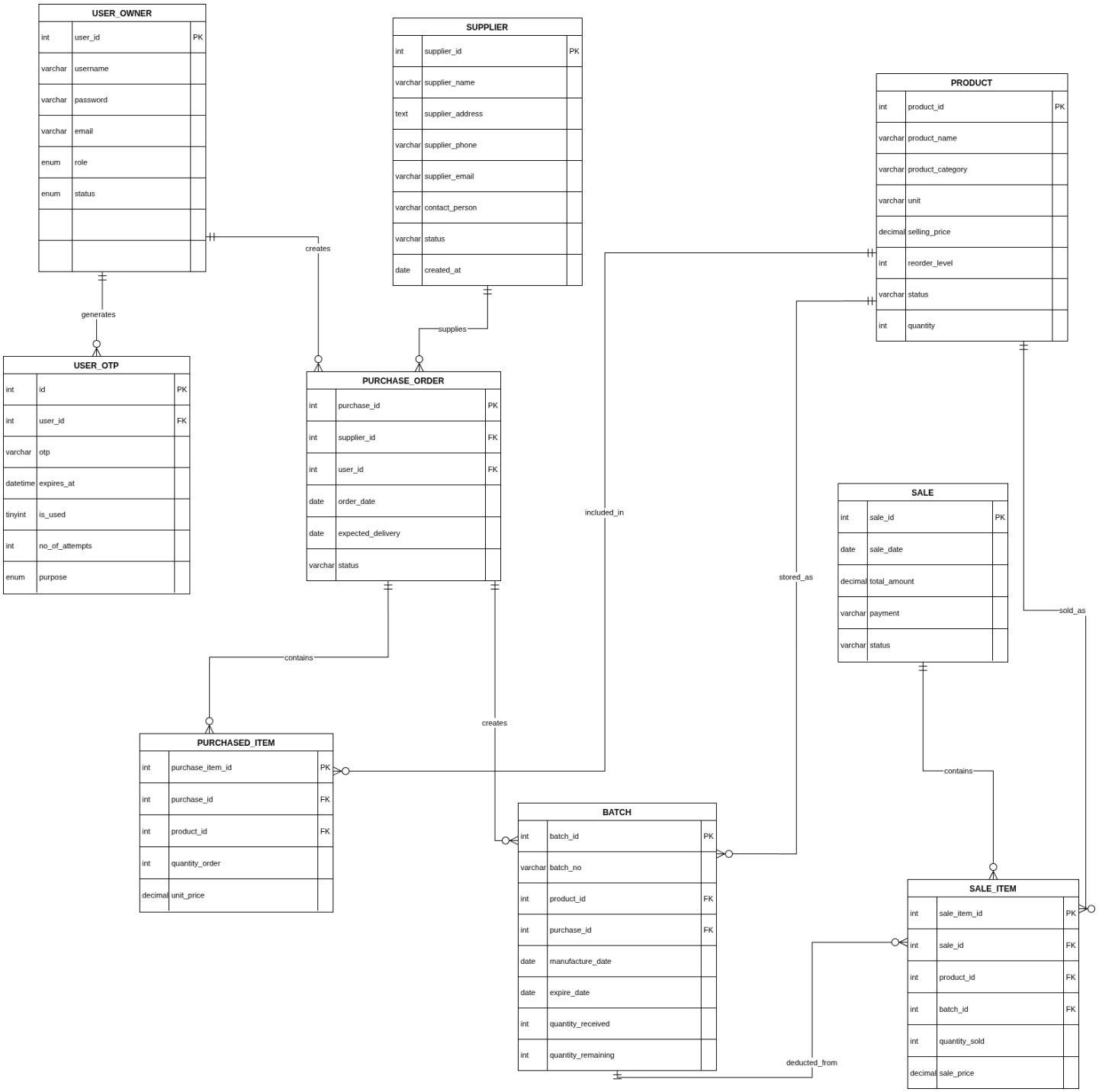
USER_OWNER
int user_id PK
varchar username
varchar password
varchar email
enum role
enum status
SUPPLIER
int supplier_id PK
varchar supplier_name
text supplier_address
varchar supplier_phone
varchar supplier_email
varchar contact_person
varchar status
date created_at
PRODUCT
int product_id PK
varchar product_name
varchar product_category
varchar unit
decimal selling_price
int reorder_level
varchar status
int quantity
creates
generates
supplies
USER_OTP
int id PK
int user_id FK
varchar otp
datetime expires_at
tinyint is_used
int no_of_attempts
enum purpose
PURCHASE_ORDER
int purchase_id PK
int supplier_id FK
int user_id FK
date order_date
date expected_delivery
varchar status
included_in
stored_as
SALE
int sale_id PK
date sale_date
decimal total_amount
varchar payment
varchar status
sold_as
contains
creates
contains
PURCHASED_ITEM
int purchase_item_id PK
int purchase_id FK
int product_id FK
int quantity_order
decimal unit_price
BATCH
int batch_id PK
varchar batch_no
int product_id FK
int purchase_id FK
date manufacture_date
date expire_date
int quantity_received
int quantity_remaining
deducted_from
SALE_ITEM
int sale_item_id PK
int sale_id FK
int product_id FK
int batch_id FK
int quantity_sold
decimal sale_price

Fig II: Entity and its descriptions

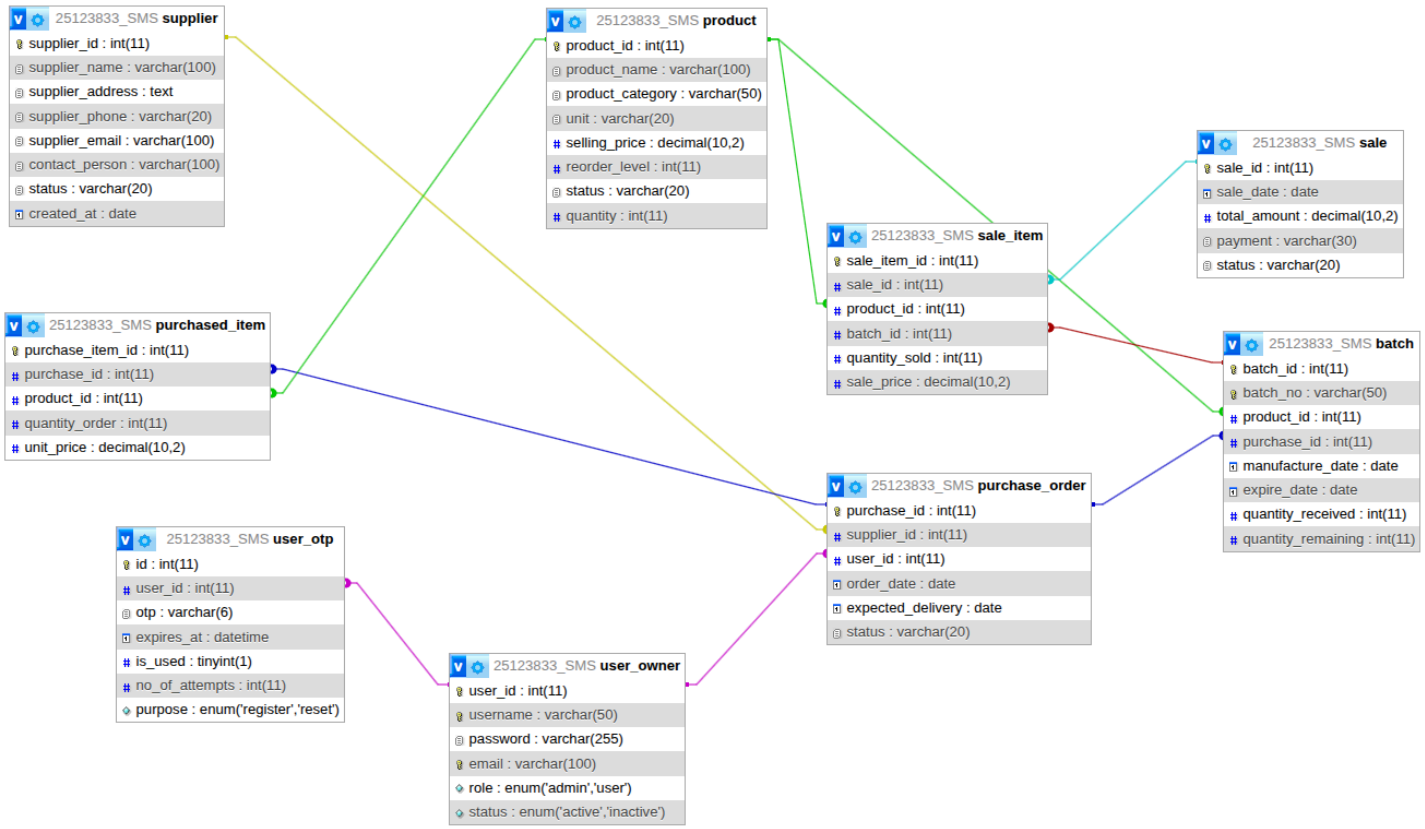

Fig III: ER diagram from the php my admin

Conclusion

Development and implementation of our Product Stock Management System is quite significant in enhancing the handling of the stock as well as managing the entire operations of the business. This system encompasses some of the vital tools such as suppliers' management, products' management, batch control, and stock control that make it a fully encompassing system in the running of the business on a day-to-day basis. For the future enhancement of the system to be useful to firms in managing their resources in an appropriate manner by making positive decisions, it is vital to continuously enhance and develop the system. The system is significantly important in bridging the gap that exists between sales and supplies by establishing an organized, reliable, and efficient working environment.